CONTOUR	# SAFETY IN DESIGN	QMS-FM035a
		Rev No. 0
		Rev Date 25/11/2021

Project Name:	**Sippy Downs Depot - Building Refurbishment**	**Client:**	**Sunshine Coast Regional Council**
Date:	**19/10/2021**	**Project Manager:**	**Paul Brennan**
Project Designer:	**Josh Truscott (Contour)**		
Risk Assessment for:	**Refurbishment of Existing Building**		

The following comprises Safety in Design (SiD) reporting for the subject project. SiD is a process defined as the integration of hazard identification and risk assessment in the design process, including provision of control strategies inherent to the design that eliminate of minimise health and safety risks for hazardous manual tasks throughout the life cycle of the project. The implementation of SiD considers safety issues associated with the concept design and detailed design of each stage of a project's life cycle or relevant to the project scope as delivered by Contour Consulting Engineers Pty Ltd, including: demolition of brownfield elements prior to construction; construction; operation; maintenance, renovation and/or modification; and demolition or demobilisation at end of life. The risk management process includes four key stages to developing and maintaining safety outcomes, including: Hazard Identification - identification of potential hazardous situations that could result in injury or illness; Risk Assessment - assessment of how likely the risk is and the associated consequence if the hazard occurs; Risk Elimination/Control - elimination or control of the risk through planned strategies and mitigation measures; and Evaluation and Review - regular review of risk controls and mitigation measures to ensure they remain current and appropriate (this process is not included in this report, as it falls outside the design phase. It is recommended that this process be included in this report, as it falls outside the design phase. It is recommended that this process be included in operational procedures in the owner/operator). The risk evaluation has been categorised in terms of a Risk Score as associated with the impact of potential consequences, and probability that the hazard will occur. The Risk Evaluation Matrix provides an Initial Risk Rating that applies if the design and/or construction and/or operations were ignorant of SiD principals and ignorant of WHS management by the Construction/Demolition/Maintenance Contractor. The risk evaluation matrix then provides a Residual Risk Rating in consideration of SiD inclusions, combined with WHS management by the Construction/Demolition/Maintenance Contractor, combined with decision-making by the Client during the design and construction phase, and combined with the due diligence of the Owner/Operator/Tenant throughout the life of the project. The SiD report is restricted to the design phase only and in no way removes or relinquishes the responsibility of the construction/maintenance/demolition/operational phase controlling entities to provide a safe working and/or operational environmental during the project life cycle.

IMPACT

Consequence	Health and Safety	To Property	To Environment	Community	Financial
Catastrophic (5)	*Fatality(ies) or permanent disability or ill-health*	*Virtual complete loss of property, plant or system*	*Long term, widespread, significant impact*	*Adverse National Media or Public Attention*	*Huge Financial Loss (>$100k)*
Major (4)	*Single death and/or long-term illness or multiple serious ??*	*Extensive damage to property, plant or system*	*Medium to long term widespread impact*	*Attention from Media or heightened concern from the Community*	*Major Financial Loss ($50k - $100k)*
Moderate (3)	*Injury; Possible hospitalisation & numerous days lost*	*Significant damage to property, plant or system*	*Moderate but reversible impact*	*Local Public or Media Attention or Complaints*	*Moderate Financial Loss ($5k - $50k)*
Minor (2)	*Minor injury; Medical treatment & some days lost*	*Damages impact on budget and program*	*Minor, short-medium term impact to local area of limited*	*Public Concern Limited to Complaints*	*Minor Financial Loss (<$5k)*
Insignificant (1)	*No or minor personal injury; First aid needed but no days lost*	*Minor damage to property, plant or system*	*Limited impact to minimal area*	*No Complaints or Concerns*	*No Financial Loss*

RISK EVALUATION
HIGH (13-25)
MEDIUM (6-12)
LOW (1-5)

IMPACT

5	**5**	**10**	**15**	**20**	**25**
4	**4**	**8**	**12**	**16**	**20**
3	**3**	**6**	**9**	**12**	**15**
2	**2**	**4**	**6**	**8**	**10**
1	**1**	**2**	**3**	**4**	**5**
Likelihood	**Rare**	**Unlikely**	**Possible**	**Likely**	**Almost Certain**
Likelihood to Occur	*Very unlikely to occur, may only occur in exceptional circumstances (1 million -1 odds or once in 100 years)*	*Unlikely to occur, not likely to occur (10000-1 or once in 10 years)*	*Likely to occur, might occur at some time (100-1 odds or once a year)*	*Good chance to occur, will probably occur in most circumstances (even odds or 10 times a year)*	*Very likely to occur, common/frequent occurrence (odds on or anytime)*

PROBABILITY

Hierarchy of Risk Control Measures	
Eliminate the Hazard	**ELIMINATE** - Get rid of the Hazard out of the workplace.
Change the Way the Work is Done	**SUBSTITUTE** - Try to replace or change plant, substances or materials to lower the risk of the hazard.
	Try to **ISOLATE** the hazard.
	ENGINEERING CONTROL - Design and install.
	ADMINISTRATIVE CONTROL - Arrange work so people spend less time around the hazard and monitor their understanding of the hazard and the controls.
Personal Protective Equipment	**PPE** - Have people wear protective equipment and clothing while near the hazard.

These Hazards apply to Demolition, Construction, Maintenance and Repair and Demolition at End of Project Life Cycle						
Risk ID	**Issue**	**Foreseeable Hazards**	**Initial Risk Score**	**Control Measures Adopted**	**Residual Risk Score**	**Action Required By**
1.1	**Access and Egress**	Access to site is via internal site road network. Potential for construction traffic to conflict with operational vehicles and personnel.	20	Contractor to develop detailed traffic management plan for the site for approval by the Client and Facility Management.	8	Contractor, Client and Facility Management
1.2		The work site is within an operational facility. Potential for operational personnel to access the work site resulting in injury.	25	Construction work to be scheduled to minimise disruption to facility operations. Recommend adequate barriers during works to prevent unauthorised access to work areas and construction travel paths.	10	Contractor, Client and Facility Management
1.3		Contractor access to the work area may conflict with facility operations. Potential for injury to operational personnel or damage to operational equipment during transport of goods to/from the work areas.	20	Construction work to be scheduled to minimise disruption to facility operations. Recommend construction travel paths are separated from facility operation travel paths, wherever possible.	10	Contractor, Client and Facility Management
2.1	**Adjoining Areas**	The work site is within an operational facility. Potential for injury to operational personnel or damage to operational equipment.	25	Construction work to be scheduled to minimise disruption to facility operations. Recommend that a detailed work plan be developed by the Contractor and approved by the Client and Facility Management prior to commencement of works.	10	Contractor, Client and Facility Management
3.1	**Electrical**	Works on or near energised electrical installations or services are defined as high risk construction work.	25	Contractor to confirm location of services prior to commencing works. Contractor to prepare a Safe Work Method Statement (SWMS) prior to commencement of works	10	Contractor
3.2		Services may be hidden, e.g. in the ground, floor or behind walls in the building. Potential for injury from contact with power.	25	Contractor to confirm location of services prior to commencing works. Contractor to prepare SWMS should services be identified. Contractor to develop and maintain isolation procedures.	10	Contractor
3.3		Contact with existing electrical installations during Demolition.	25	Contractor to confirm location of services prior to commencing works. Contractor to prepare SWMS should services be identified. Contractor to develop and maintain isolation procedures.	10	Contractor
4.1		The work site is directly adjacent to a Diesel Refilling Station. Access will need to be maintained for refilling operations during construction activities. This will create potential conflicts between operational staff and equipment and construction staff and equipment.	25	Construction and Demolition work to be scheduled to minimise disruption to facility operations. Recommend that a detailed work plan be developed by the Contractor and approved by the Client and Facility Management prior to commencement of works. Contractor must prepare a SWMS prior to commencing works.	10	Contractor, Client and Facility Management

4.2	**On-Site Fuel Refilling Station**	The work site is directly adjacent to a Diesel Refilling Station. Demolition and constructoin activities will need to be cognisant of the proximity of the Diesel Refilling Station to the worksite.	25	Construction and Demolition work to be scheduled to minimise disruption to facility operations. Recommend that a detailed work plan be developed by the Contractor and approved by the Client and Facility Management prior to commencement of works. Contractor must prepare a SWMS prior to commencing works.	10	Contractor, Client and Facility Management
4.3		Mobile plant equipment operating near Refilling Station.	25	Contractor to prepare SWMS and ensure mobile plant operators are fully inducted before commencement of services.	10	Contractor
5.1	**Mobile Plant**	Work carried out where there is movement of powered mobile plant is considered a high risk construction activity.	25	Contractor must prepare a SWMS prior to commencing works.	10	Contractor
5.2		The work site is within an operational facility. Potential for injury to operational personnel or damage to operational equipment by mobile plant.	25	Construction and Demolition work to be scheduled to minimise disruption to facility operations. Recommend that a detailed work plan be developed by the Contractor and approved by the Client and Facility Management prior to commencement of works.	10	Contractor, Client and Facility Management
5.3		Mobile plant equipment operating on or near service pits.	20	Contractor to identify service pits and provide details to mobile plant operators.	8	Contractor
6.1	**Noise**	Noise from works affecting operational personnel in adjacent areas.	10	Contractor and Client to inform operatonal personnel of expected noise levels and times when loud noise will be generated from the site.	4	Contractor, Client and Facility Management
7.1	**Hidden Services**	Contact with hidden services, including electricity, gas, telecommunications, sewer, water and stormwater.	25	Contractor to confirm location of services prior to commencing works. Contractor to prepare SWMS should services be identified.	10	Contractor
8.1	**Hazardous Substances**	Cutting of masonry blocks. Potential inhalation of silica.	25	Recommend minimising dust exposure as follows: - avoid cutting where possible via use of pre-cut blocks. - work in well-ventilated areas. - utilise appropriate PPE. Follow directions of product SDS.	10	Contractor
8.2		Treated timber used in construction.	10	Follow directions of product SDS. Utilise appropriate PPE.	4	Contractor
8.3		Fibre cement used in construction - inhalation of silica dust and contact with sealant (fibre cement selected for durability and low maintenance requirements).	25	Follow directions of product SDS. Work in well ventilated area and utilise appropriate PPE.	10	Contractor
8.4		VOC fumes from paints and other coatings.	20	Recommend use of zero VOC paints where possible. Follow directions of product SDS. Utilise appropriate PPE.	8	Contractor

8.5		VOC fumes from internal joinery	20	Recommend use of zero or low emission materials for joinery. Follow directions of product SDS. Utilise appropriate PPE.	8	Contractor
8.6		Reticulated gas and lines. Damage to pipes and/or gas leaks leading to inhalation or explosion hazards.	25	Contractor to confirm location of services prior to commencing works. Contractor to prepare SWMS should services be identified. Contractor to develop and maintain isolation procedures.	10	Contractor
9.1	**Structural Stability**	Alteration to load bearing structures. Potential collapse and weakening of the existing structure during demolition.	25	Structural elements to be demolished are considered high risk construction work and Contractor to prepare a SWMS prior to commencement of works. Where necessary, Contractor to provide propping and support of key structural elements during demolition process.	10	Contractor
10.1	**Structural Integrity**	Structural Failure.	25	Design structural elements in accordance with relevant standards and guidelines.	5	Designer
11.1	**Working at Height**	Work at height is considered to be a high risk construction activity. Demolition and construction of walls and roofing..	25	Contractor to prepare a SWMS prior to commencement of works.	10	Contractor
12.1	**Working in Confined Spaces**	Working in confined space is considered to be a high-risk construction activity. Demolition and construction of walls and roofing.	25	Recommend the use of PPE - refer AS2865. Contractor to prepare a SWMS prior to commencement of works.	10	Contractor
13.1	**Falling Objects**	Falling debris during construction and demolition. Falling tools and/or materials from heights.	25	Recommend the use of PPE - refer AS2865. Contractor to prepare a SWMS prior to commencement of works.	10	Contractor

OPERATION PHASE - These Hazards apply to Use of the Building for which it is Designed

Risk ID	Issue	Foreseeable Hazards	Initial Risk Score	Control Measures Adopted	Residual Risk Score	Action
1.1	**Access and Egress to Storage Rooms**	Access for persons with disabilities.	10	PWD access to comply with AS1428	2	Designer and Contractor
1.2		Pedestrian trips and falls.	10	Walkways, ramps and stairs to comply with AS1428	2	Designer and Contractor

2.1	**Access and Egress to Toilet**	Pedestrian trips and falls.	**10**	Walkways, ramps and stairs to comply withNCC 2019	**2**	Designer and Contractor
2.2		Access for persons with disabilities.	**10**	This toilet is not suitable for PWD access, we consdier the facility has suitable PWD accessible amenities located at the front of the complex.	**2**	Client
3.1	**Lighting**	Inadequate lighting to toilets and storage areas.	**15**	Lighting to comply with AS1680.	**3**	Designer, Contractor and Facility Management
3.2		Inadequate emergency lighting	**10**	Standards do not require Emergency lighting, notwithstanding we consider it should be installed given the intended use of the facilities.	**2**	Designer, Contractor and Facility Management
4.1	**Slips, Trips and Falls**	Slipping on finished floor surfaces.	**15**	Slip resistant R11/P4 floor surfaces throughout all wet areas	**9**	Designer and Contractor
4.2		Slipping on floor surfaces due to poor cleaning and maintenance.	**15**	Facility management to implement appropriate cleaning and maintenance regimes.	**9**	Facility Management
5.1	**Fire**	The refurbished building is within 8.0m of the Diesel Refilling Station.	**25**	Given the building is pre-existing and pre-dated the adjancent Diesel Refilling Station, and our works relates to refurbishment only, we consdier the design/construction of the Diesel Refilling Station was facilitated in accordance with AS1940. We note that the Building is approxmiately 8.0m from the Diesel Refilling Point	**10**	Client and Facility Management
6.1	**Regular Inspection and Maintenance**	Interaction with aging built infrastructure.	**12**	Facility Management is to be vigilant and is to undertake regular maintenance of infrastructure and signage as may be required to ensure safe operation in accordance with community expectations, the installed design features, and/or associated design documentation.	**3**	Facility Management
7.1	**Current and Appropriate Infrastructure**	Interaction with aging built infrastructure.	**12**	Facility Mangement is to be vigilant and to undertake regular review of risk controls and mitigation measures to ensure they remain current and appropriate.	**3**	Facility Management